

Find the Mistake: Signs and Denominators in Conic Equations

- Every problem quotes a value someone actually wrote. Produce the correct value, and where asked, name the move that produced the wrong one.
 - Standard forms are written with subtractions: $(x - h)^2 + (y - k)^2 = r^2$. A plus inside a bracket means a negative coordinate.
 - The right-hand side of a circle equation is r^2 , and the numbers under x^2 and y^2 in an ellipse or hyperbola are a^2 and b^2 — squares, every one of them.
 - Foci: *subtract* for an ellipse, $c^2 = a^2 - b^2$; *add* for a hyperbola, $c^2 = a^2 + b^2$.
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1. For the circle $(x + 5)^2 + (y - 2)^2 = 36$, a student reports the centre as $(5, -2)$.

(a) Give the correct x -coordinate of the centre.

Answer: _____

(b) Explain what about the standard form makes that coordinate negative.

Answer: _____

2. For the circle $(x - 1)^2 + (y + 4)^2 = 49$, a student reports the radius as 49. Find the correct radius.

Answer: _____

3. For the ellipse $\frac{x^2}{9} + \frac{y^2}{25} = 1$, a student says the semi-major axis has length 3.

(a) Find the correct length of the semi-major axis.

Answer: _____

(b) Explain the error, and say which way the major axis actually runs.

Answer: _____

4. For the ellipse $\frac{x^2}{169} + \frac{y^2}{25} = 1$, a student computes the distance from the centre to a focus as $\sqrt{169 + 25}$, giving about 13.93. Find the correct distance from the centre to a focus.

Answer: _____

5. For the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, a student computes the distance from the centre to a focus as $\sqrt{16 - 9}$, giving about 2.65. Find the correct distance from the centre to a focus.

Answer: _____

6. The circle $x^2 + y^2 + 6x - 8y - 11 = 0$ is to be put into standard form. A student reports the centre as $(6, -8)$.

(a) Find the correct centre.

Answer: _____

(b) Find the radius.

Answer: _____

(c) Explain the two separate mistakes in the quoted centre.

Answer: _____

7. For the hyperbola $\frac{y^2}{36} - \frac{x^2}{64} = 1$, a student gives the asymptote slopes as $\pm \frac{8}{6}$. Find the correct positive asymptote slope.

Answer: _____

8. The circle $x^2 + y^2 - 4x + 10y + 20 = 0$ is handed to two students. The first says the radius is $\sqrt{20}$, about 4.47. The second says the centre is $(2, 5)$.

(a) Find the correct radius.

Answer: _____

(b) Find the correct y -coordinate of the centre.

Answer: _____

(c) Explain the two errors separately: one for each student.

Answer: _____